

TOPO-BATHY LIDAR & IMAGERY POST SURVEY REPORT

‘Eua, Vava’u, Tafahi & Niuatoputapu
Tonga

CONTRACT No.: SPC 20-293

PREPARED FOR:
THE PACIFIC COMMUNITY (SPC)

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1. INTRODUCTION

Woolpert Inc. was contracted by the The Pacific Community (SPC) to provide airborne topographic and bathymetric lidar acquisition and processing services, along with aerial imagery, for selected areas within Tonga, specifically the islands of 'Eua, Vava'u, Tafahi & Niuatoputapu.

All data were acquired using Woolpert's Leica HawkEye 4X system to meet the requirements of the project.

Details of the survey, data processing, QC and product creation are provided in detail within this report.

1.1. SURVEY AREA

The survey extents provided by SPC for Tonga covered approximately 1001 square kilometers. Survey areas covered the islands of 'Eua, Vava'u, Tafahi & Niuatoputapu. Woolpert worked with the SPC to adjust the Niuatoputapu boundary provided in the statement of work to expand the area to increase data collection.

The project area was split into survey blocks, allowing flight lines to be planned in the most efficient manner. Lidar data were collected to support 100% coverage.

Table 1: Survey Blocks

Area	Area (SqKm)	Survey Block(s)
'Eua	188	BL30, BL32, BL33, BL34
Vava'u	730	BL50, BL51, BL52
Tafahi	9	BL41, BL42, BL43, BL44, BL45
Niuatoputapu	74	BL40

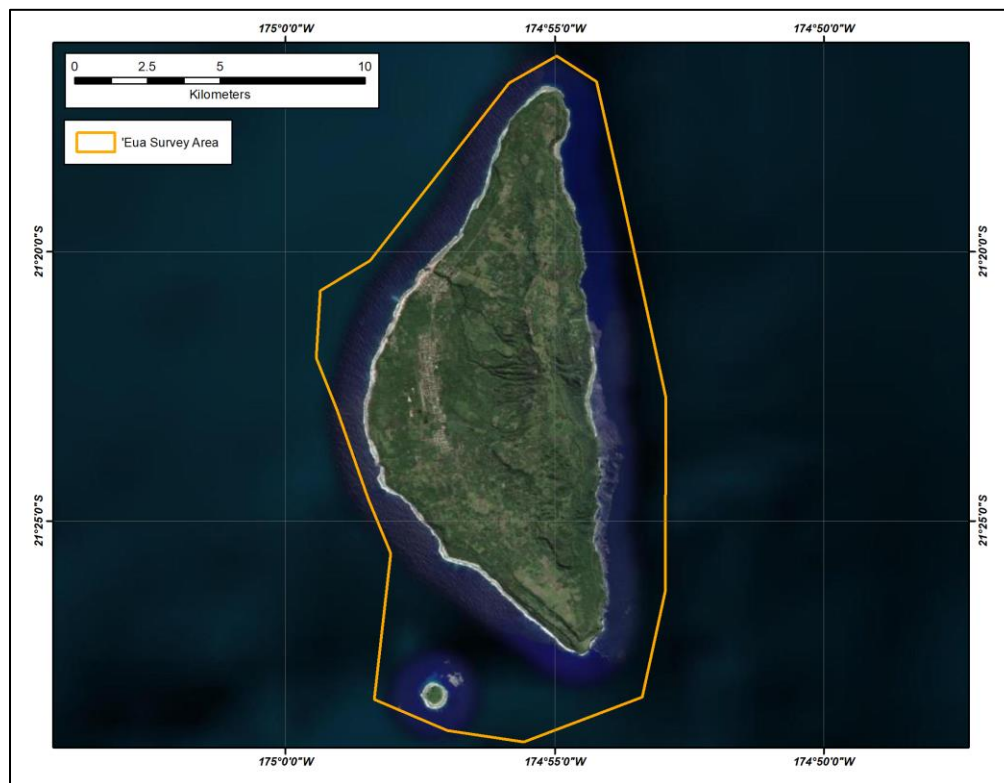


Figure 1: 'Eua Survey Area

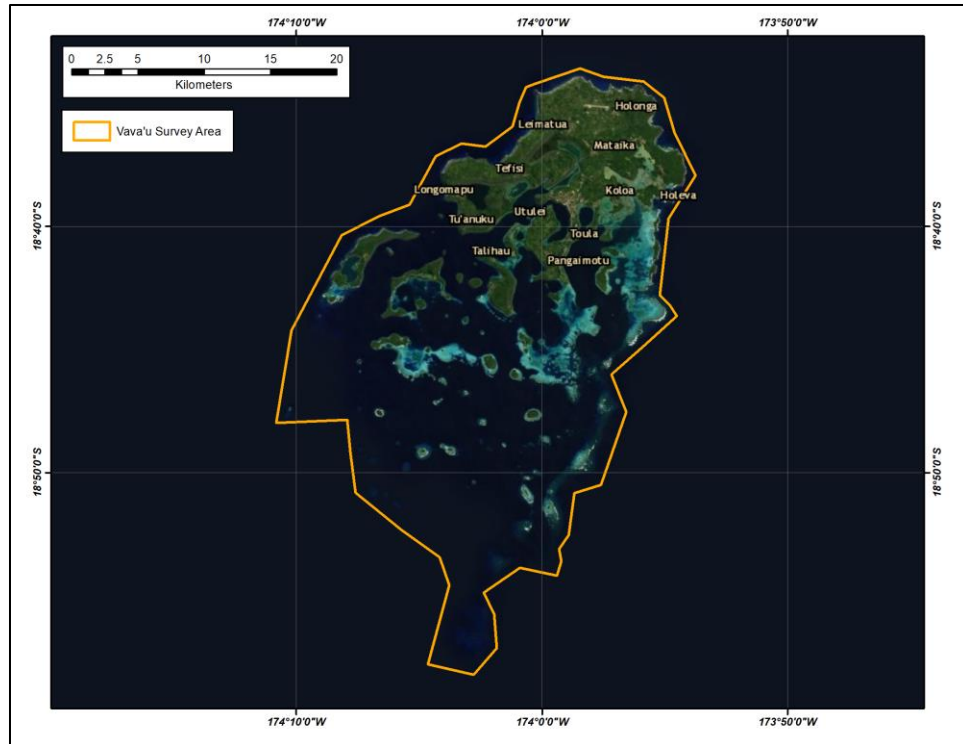


Figure 2: Vava'u Survey Area

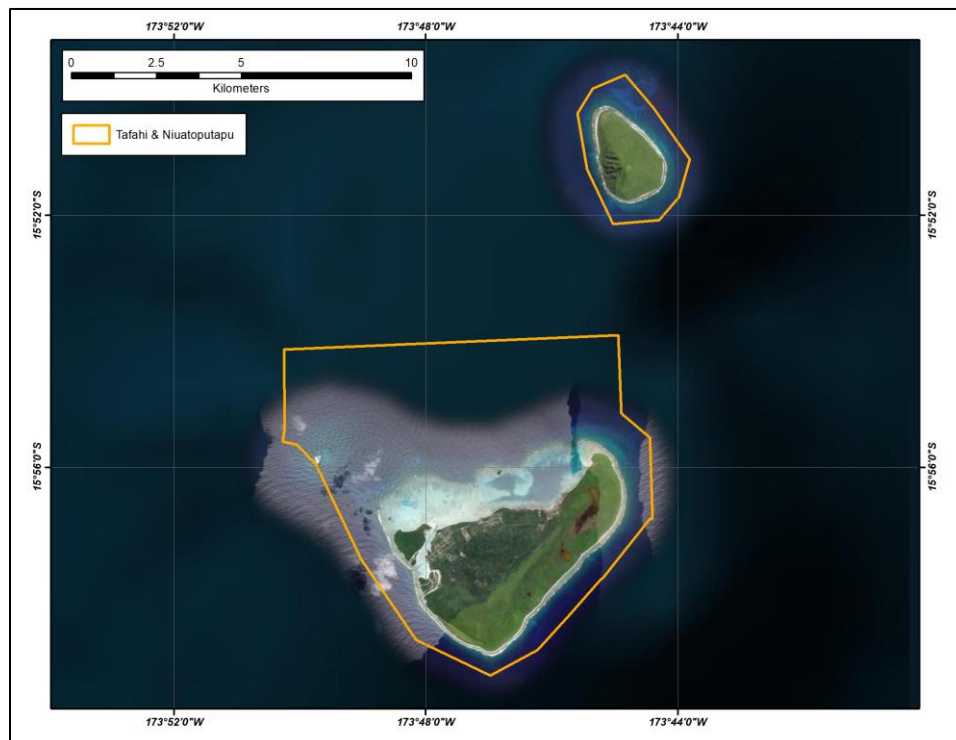


Figure 3: Tafahi & Niutopotupu Survey Area

2. LIDAR DATA ACQUISITION

All lidar data were acquired using a Chiroptera 4X (CH4X) sensor, with an additional Leica 40kHz deep bathymetric channel. The combination of these sensors is referred to as a Leica HawkEye 4X (HE4X) system. The CH4X sensor was mounted in a Leica PAV100 gyro-stabilized mount integrated with a NovAtel SPAN GNSS and LCI-100C IMU. The deep channel sensor head was mounted over a second hatch, with an additional IMU. Real time navigation and GNSS/IMU data logging was provided by Leica FlightPro software. Lidar data were logged on the Airborne Hydrography, AB (AHAB) operator console.

The aircraft was integrated with the sensor prior to this project. The aircraft arrived in Tongatapu, Tonga on 12 May 2022, where the crew underwent a 7-day quarantine period. A calibration flight was conducted at Fua'amotu International Airport (TBU) on 20 May 2022. The first mobilization collection of project data has been acquired over 18 flights from 20 May 2022 to 30 June 2022.

2.1. MOBILIZATION

The HE4X sensor was installed in a Reims-Cessna 406 (ZK-XLF) owned and operated by Kiwi Air (Figure 4).



Figure 4: Mobilized Aircraft

2.1.1. AIRCRAFT OFFSET SURVEY

Physical mounting offsets between the GNSS antenna, IMU and gyro-stabilized mount were determined through a combination of manual measurements and iterative processing in NovAtel Inertial Explorer software.

Manual measurements were taken from the GNSS antenna to the reference point on the CH4X sensor and the deep channel sensor head. These measurements are combined with the known offsets between the reference point, IMU, and in the case of the CH4X sensor head, the rotation center of the gyro-stabilized mount. This preliminary value was then used to seed the post-processing software which, through an iterative computation, uses the dynamic accelerations and rotations during flight to refine the offsets. Once the solution converges, the final offsets are entered into the flight management software and used in subsequent post-processing of the GNSS/IMU data for final trajectories.

Final offsets, shown in the Leica reference frame, are presented in Table 2.

Table 2: Aircraft Offsets

Sensor Head	Lever Arm	X (forward)	Y (right)	Z (down)
CHII (Topo and Shallow Channel)	Reference to GNSS Antenna L1 Phase Center	0.604 m	-0.056 m	-1.226 m
	Reference to IMU	-0.003 m	-0.006 m	-0.296 m
	Reference to IMU Rotation	0 °	0 °	90 °
Deep Channel	IMU to GNSS Antenna L1 Phase Center	1.234	-0.134	-0.825
	Reference to IMU Rotation	0°	90°	-90°

2.2. CALIBRATION

Field calibration of the HE4X system is carried out to eliminate systematic errors by calculating corrections for boresight errors, scanner angle errors, remaining IMU angle errors and any necessary internal timing errors. To verify or compute the field calibration, the following lines are flown (Figure 5):

- 2 x Line A over mixed terrain with flat or gentle slopes and features such as peaked roof buildings (1 x each direction)
- 1 x Line B offset +50% from Line A in one direction
- 1 x Line C offset -50% from Line A in the same direction as Line B
- 2 x Line D orthogonal to previous lines (1 x each direction)

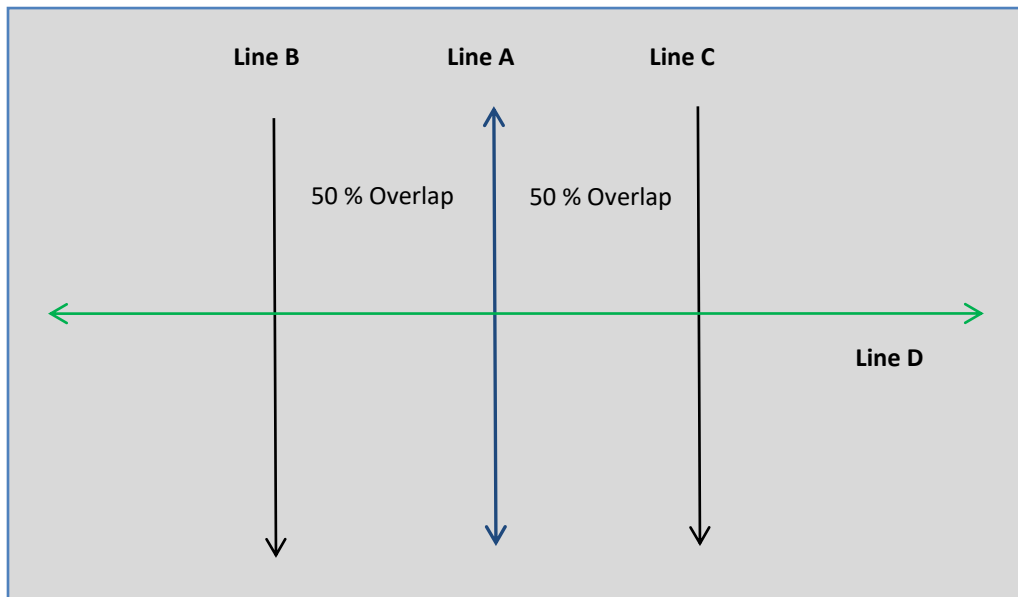


Figure 5: Schematic of HE4X Calibration Lines

A set of calibration lines were acquired at 600m, 500m, and 400m altitude. All sets of lines are used to calibrate and verify the topographic lidar, while the 500m and 400m lines are used for the bathymetric lidar.

Calibration values are calculated using the automatic calibration routine within the Leica Lidar Survey Studio (LSS) software. This utility first identifies patches or areas of gentle slope within the overlap region of all the lines to use for calibration. Patch selection prevents areas of vegetation, side of cars or buildings, from being used in the calibration process. Next, the utility compares the front side and back side of the elliptical scan within the same line, as well as comparing all lines to each other, to identify suitable calibration parameters such that data within the patches match. The procedure is iterative and continues until the best possible solution is computed.

Calibration for each channel (topo, shallow and deep) is done independently. Topo channel calibration was computed using the 500m altitude lines. The 600m and 400m lines were then used for verification. Calibration of

the shallow channel and deep channel were computed using 500m altitude data. Any lower altitude data were used for verification.

At each step of the calibration process, quality assurance is conducted to ensure values being calculated are valid. This is done using the Leica LSS Quality Control Utility. Two types of checks are done; firstly, the front scan is compared to the back scan for every line. Secondly, a single line is chosen as a baseline and is compared to every other line. We would expect the average errors from both of these checks to be small. In addition, the data is visually reviewed. In particular, features are studied to ensure lines from different directions show structures in the same position, in other words, verifying horizontal accuracy is maintained. These tests all provide assurance of relative accuracy.

Results from the calibration verification checks are provided in Table 3 below. Results are good and indicate that calibration was successful. Calibration values computed were used for the entire project.

Table 3: Calibration QA Results

Test		Topo 600m	Topo 500m	Topo 400m	Shallow 500m	Shallow 400m	Deep 500m	Deep 400m
Front to Back Scan Comparison	Average Error (m)	-0.0024	0.0012	-0.0006	0.0012	0.0006	-0.0002	0.0040
	Std. Dev. of Error	0.0006	0.0004	0.0004	0.0006	0.0008	0.0048	0.0069
Line to Line Comparison	Average Error (m)	-0.0032	0.0044	0.0108	0.0054	0.0113	0.0026	0.0119
	Std. Dev. of Error	0.0018	0.0008	0.0013	0.0029	0.0017	0.0082	0.0089

Woolpert acquired a detailed set of ground truth data over the apron at the Fua'amotu International Airport. The ground truth was acquired using Trimble R10 GNSS receivers, Real Time Kinematic (RTK) and Post-Processed Kinematic (PPK) survey techniques.

Ground truth is not used within the automatic calibration routine; however, a comparison to the lidar data was used to verify absolute accuracy. Results presented in Table 4 show data is well within required accuracy specifications.

Table 4: Calibration Ground Truth Comparisons

	Topo	Shallow	Deep
Average dz (m)	0.0003	0.0118	0.0004
St Dev (m)	0.0071	0.0028	0.0015

2.3. SURVEY OPERATIONS

For ease of operations and data management, the Tonga survey area was split into survey blocks. Actual flight lines flown, including start and end date and unique line ID, are provided in the flight line database included with the project deliverables in SHP format.

A summary of the daily operations is shown in Table 5. Detailed Flight Logs for each day are provided in Appendix A. Since the survey areas covered a large area two locations were used to base operations. The collection of 'Eua was based out of the Fua'amotu International Airport. The collection of Vava'u, Tafahi and Niuatoputapu was based out of the Vava'u International Airport.

Six cross lines were acquired across the during the survey for quality purposes. Crosslines were planned perpendicular to main scheme survey lines and used to verify the relative accuracy of the data where bottom coverage allowed.

Table 5: Summary of Daily Operations

Flight	Engine Time	Air Time	Flown km %		Reflown km %		Comments
2022-05-20A	4:15:00	1:50:00	108.8	1.9%	10.5	0.2%	Calibration, 'Eua
2022-05-21A	3:02:00	2:20:00	251.2	4.5%	22.3	0.4%	'Eua
2022-05-22A	5:59:00	5:23:00	494.6	8.8%			'Eua
2022-05-24A	2:17:00	1:38:00	139.8	2.5%	36.7	0.7%	'Eua
2022-06-02A	4:19:00	3:40:00	98.9	1.8%	15.1	0.3%	'Eua
2022-06-06A	4:38:00	3:53:00	237.0	4.2%	246.8	4.4%	'Eua
2022-06-07A	3:04:00	2:16:00	124.6	2.2%	9.7	0.2%	'Eua
2022-06-09A	5:38:00	5:03:00	804.6	14.3%	48.0	0.9%	Vava'u
2022-06-10A	5:18:00	4:42:00	656.6	11.6%	35.4	0.6%	Vava'u
2022-06-12A	6:12:00	5:44:00	892.2	15.8%	7.7	0.1%	Vava'u
2022-06-13A	6:08:00	5:34:00	899.3	16.0%	115.5	2.0%	Vava'u
2022-06-14A	4:00:00	3:26:00	550.2	9.8%			Vava'u
2022-06-19A	2:27:00	1:58:00	6.4	0.1%	278.6	4.9%	Vava'u
2022-06-21A	5:40:00	4:54:00	338.8	6.0%	2.3	0.0%	Tafahi, Niutopotapu
2022-06-27A	3:34:00	3:02:00	70.8	1.3%			Tafahi
2022-06-28A	4:37:00	4:00:00	147.5	2.6%	24.9	0.4%	Vava'u
2022-06-29A	4:56:00	4:18:00	162.6	2.9%	22.3	0.4%	Vava'u, Tafahi, Niutopotapu
2022-06-30A	3:45:00	3:18:00	19.3	0.3%	107.7	1.9%	Tafahi, Niutopotapu
TOTAL	79:49	66:59	6003	106.5%	983	17.4%	

2.3.1. THE HAWKEYE 4X

All lidar data were acquired using a HE4X sensor. The HE4X is a latest generation topographic and bathymetric lidar sensor. The system provides denser data than previous traditional bathymetric lidar systems. It is unique in its ability to acquire bathymetric lidar, topographic lidar and 4-band digital camera imagery simultaneously.

The HE4X provided 500 kHz topographic data, an effective 140 kHz shallow bathymetric data and 40 kHz deep bathymetric data. 4-band 80 MP digital camera imagery was also collected simultaneously with the sensor's RCD-30 camera.

The bathymetric and topographic lasers are independent and do not share an optical chain or receivers, so they are optimized for their specific function. As with any bathymetric lidar, maximum depth penetration is a function of water clarity and seabed reflectivity. The HE4X is designed to penetrate to 3 times the secchi depth. This is also represented as $D_{max} = 4/K$, where K is the diffuse attenuation coefficient, and assuming K is between 0.1 and 0.3, a normal sea state and 15% seabed reflectance.

Both the topographic and bathymetric sub-systems use a palmer scanner to produce an elliptical scan pattern of laser points with a degree of incidence ranging from +/-14° (front and back) to +/-20° (sides), providing a 40° field of view. This has the benefit of providing multiple look angles on a single pass and helps to eliminate shadowing effects. This can be of particular use in urban areas, where all sides of a building are illuminated, or for bathymetric features such as the sides of narrow water channels, or features on the seafloor such as smaller objects and wrecks. It also

assists with penetration in the surf zone where the back scan passes the same ground location a couple of seconds after the front scan, allowing the areas of whitewater to shift.

The bathymetric laser is a diode pumped class 4 laser which operates in the green spectrum. Full waveform data is acquired for every pulse. The topographic laser operates in the infra-red spectrum at 1064nm. Up to 4 returns per pulse are acquired from each lidar.

For this project, the flight parameters shown in Table 6 were used to provide 100% coverage. Flight parameters used exceed the requirements for the survey to meet IHO Order 1B.

Table 6: HE4X Survey Flight Parameters

	Topo-Bathy Flight Lines	Topo Only Flight Lines	
Topo PRF (kHz)	300	440	500
Topo Points per m ²	>12	>10	>12
Shallow Bathy PRF (kHz)	300		
Shallow Bathy Points per m ²	5.6	7.2	5.6
Deep Bathy PRF (kHz)	40	N/A	N/A
Deep Bathy Points per m ²	1.3	N/A	N/A
Swath Width (m)	290	450	290
Flight Line Sidelap (%)	20		
Altitude (m)	400	620	400
Survey Speed (knots)	125		

During acquisition, flight lines are shown on a pilot's display, and the aircraft is controlled by the pilot at all times. The HE4X system includes a NovAtel SPAN GNSS system with an LCI-100C IMU for aircraft position and orientation. One IMU is in the main Chiroptera sensor head, which includes the topo channel, shallow channel and RCD30 camera. Data from this IMU are also used in real-time by the PAV100 gyro-stabilized mount to compensate for deviations in pitch and roll. A second IMU is contained within the deep channel sensor head, installed over a second hatch in the aircraft. This head does not include a gyro-stabilized mount. Aircraft bank angles were restricted to 25° to avoid any potential GNSS dropouts. No flights were planned if the PDOP was expected to go above 3.0.

Data were monitored for quality during acquisition using the Operators Console running on the AHAB collection computer. The operator monitored system status of the scanners and receivers, waveforms, camera images, data coverage, flight lines and the health of the navigation system.

All data were recorded to a removable solid-state hard disk. At the end of the flight the hard disk was removed and taken to the field office where data were copied onto backup disks for transmittal back to the main processing office. Preliminary data were reviewed daily in the field for quality and coverage.

2.3.2. POSITIONING

Position and orientation data were acquired in the aircraft using a NovAtel SPAN with LCI-100C IMU. All data were post-processed using NovAtel Inertial Explorer software to provide a tightly coupled position and orientation trajectory solution.

Project areas 'Eua and Vava'u were processed using single-base Post Processed Kinematic (PPK) techniques. GNSS base stations were set up at base operating airports Fua'amotu International Airport and Vava'u International Airport. The data collected from GNSS base stations provided the reference station data used to control trajectory processing.

Final coordinates for TBU1 and VAV1 were uploaded and computed by Trimble's RTX Post-Processing services, with daily results averaged to finalize the coordinates.

Table 7: GNSS Base Summary

GNSS Base Station	Latitude	Longitude	Height (m)	Datum
TBU1	21° 14' 40.88798" S	175° 08' 16.48436" W	85.703	ITRF2014 (Epoch 2022.38)
VAV1	18° 35' 08.08803" S	173° 58' 03.37296" W	115.050	ITRF2014 (Epoch 2022.44)

Tafari and Niuatoputapu data were acquired using Precise Point Positioning (PPP) techniques and no GNSS base stations were used. Logs for the base station and trajectory processing are provided in Appendix B.

3. SURVEY CHECK POINTS

For quality control purposes a daily QC flight line was acquired over the same location every day. Two QC flight lines were used for operations, one flown when flying from the Fua'amotu International Airport and the other flown from the Vava'u International Airport.

Topographic control point data will be collected in the second mobilization to Tonga. The control points will be collected using RTK GNSS techniques over the area covered by the daily lidar QC lines.

Further discussion of the processing and results will be presented in the Final Project Report.

APPENDIX A : FLIGHT LOGS

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / 400m, 500m, 600m, BL32	DATE:	20 May 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.

MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 2400ft
BASE STATION:	TBU1	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04

ENGINE START:	20:00	ENGINE OFF:	0:15	ENGINE TIME:	04:15
TAKEOFF:	22:08	LANDING:	23:58	AIR TIME	01:50

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:08:00				Takeoff
		22:13:39				DS: 600m_20220520_221339
000_FL1	6361	22:13:39	22:15:15	600	440 13	
001_FL2	6362	22:17:36	22:19:16	600	440 13	
002_FL3	6363	22:21:51	22:23:33	600	440 13	
003_FL4	6364	22:25:40	22:27:18	600	440 13	
004_FL5	6365	22:29:09	22:30:55	600	440 13	
005_FL6	6366	22:35:35	22:37:17	600	440 13	
		22:39:53				DS: 500m_20220520_223953
000_FL1	6351	22:39:53	22:41:24	480	500 13	
001_FL2	6352	22:43:36	22:45:10	480	500 13	
002_FL3	6353	22:47:23	22:48:52	480	500 13	
003_FL4	6354	22:50:59	22:52:27	480	500 13	
004_FL5	6355	23:00:25	23:01:53	480	500 13	
005_FL6	6356	23:05:47	23:07:18	480	500 13	
		23:09:37				DS: 400m_20220520_230937
000_FL1	6341	23:09:37	23:11:05	400	300 10	
001_FL2	6342	23:13:23	23:14:49	400	300 10	
002_FL3	6343	23:17:17	23:18:41	400	300 10	
003_FL4	6344	23:28:29	23:29:51	400	300 10	
004_FL5	6345	23:31:41	23:33:08	400	300 10	
005_FL6	6346	23:36:58	23:38:22	400	300 10	
		23:42:52				DS: BL32_20220520_234252
000_FL20	3295	23:42:52	23:45:39	400	300 10	BAD: Incorrect Power settings
		23:50:00				RTB
		23:58:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL30, BL31, BL32, BL33	DATE:	21 May 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua	CLOUDS:	Clouds @ 2400ft
BASE STATION:	TBU1	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-04
ENGINE START:	0:40	ENGINE OFF:	3:42
TAKEOFF:	1:08	LANDING:	3:28
		ENGINE TIME:	03:02
		AIR TIME	02:20

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:08:00				Takeoff
		01:12:30				DS: BL32_20220521_011230
000_FL20	3295	01:12:30	01:13:03	400	300 12	BAD: Problem with chiller
001_FL20	3295	01:17:58	01:19:54	400	300 12	
		01:22:22				DS: BL31_20220521_012222
000_FL34	3195	01:22:22	01:22:33	400	300 12	BAD: Eyesafety triggered by terrain
001_FL34	3195	01:27:06	01:30:14	400	300 12	
		01:34:32				DS: BL30_20220521_013432
000_FL17	3095	01:34:32	01:36:25	400	300 12	
		01:37:00				All 3 XL collected moving to BL33
		01:42:11				DS: BL33_20220521_014211
000_FL31	3302	01:42:11	01:47:19	620	400 14	
001_FL30	3301	01:49:29	01:54:56	620	400 14	
002_FL29	3329	01:57:19	02:02:46	620	400 14	
003_FL28	3328	02:04:30	02:10:07	620	400 14	
004_FL27	3327	02:12:02	02:17:45	620	400 14	
005_FL26	3326	02:19:42	02:25:20	620	400 14	
006_FL25	3325	02:27:14	02:32:47	620	400 14	
007_FL24	3324	02:34:40	02:40:10	620	400 14	
008_FL23	3323	02:42:01	02:47:26	620	400 14	
009_FL22	3322	02:49:27	02:54:49	620	400 14	
010_FL21	3321	02:56:28	03:02:00	620	400 14	
011_FL20	3320	03:03:38	03:08:42	620	400 14	
012_FL19	3319	03:10:46	03:15:44	620	400 14	
013_FL18	3318	03:17:23	03:22:09	620	400 14	
		03:28:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL31, BL32, QC61	DATE:	22 May 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 1500ft
BASE STATION:	TBU1	WIND:	10-15 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	20:08	ENGINE OFF:	2:07
TAKEOFF:	20:33	LANDING:	1:56
		ENGINE TIME:	05:59
		AIR TIME	05:23

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:33:00				Takeoff
		20:36:23				DS: QC61_20220522_203623
000_FL1	6101	20:36:23	20:37:54	400	300 12	
		20:52:14				DS: BL31_20220522_205214
000_FL33	3133	20:52:14	20:56:16	400	300 12	
001_FL32	3132	20:58:02	21:02:19	400	300 12	
002_FL31	3131	21:04:20	21:08:33	400	300 10	
003_FL30	3130	21:10:24	21:14:38	400	300 10	
004_FL29	3129	21:16:26	21:20:38	400	300 10	
005_FL28	3128	21:22:23	21:26:34	400	300 10	
006_FL27	3127	21:28:21	21:32:36	400	300 10	
007_FL26	3126	21:34:20	21:38:35	400	300 10	
008_FL25	3125	21:40:20	21:44:27	400	300 10	
009_FL24	3124	21:49:11	21:53:09	400	300 10	
010_FL23	3123	21:55:23	21:59:08	400	300 10	
011_FL22	3122	22:00:50	22:04:32	400	300 10	
012_FL21	3121	22:06:32	22:09:59	400	300 10	
013_FL20	3120	22:31:42	22:35:08	400	300 10	
014_FL19	3119	22:39:42	22:42:49	400	300 10	
015_FL18	3118	22:45:00	22:48:03	400	300 10	
016_FL17	3117	22:49:59	22:52:48	400	300 10	
017_FL16	3116	23:08:05	23:10:54	400	300 10	
018_FL15	3115	23:12:52	23:15:33	400	300 10	
019_FL14	3114	23:17:29	23:20:06	400	300 10	
020_FL13	3113	23:21:44	23:24:16	400	300 10	
021_FL12	3112	23:26:24	23:28:49	400	300 10	
022_FL11	3111	23:30:21	23:32:45	400	300 10	
023_FL10	3110	23:34:37	23:36:52	400	300 10	
024_FL9	3109	23:38:30	23:40:39	400	300 10	
025_FL8	3108	23:42:32	23:44:35	400	300 10	
026_FL7	3107	23:46:13	23:48:08	400	300 10	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL31, BL32, QC61	DATE:	22 May 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 1500ft
BASE STATION:	TBU1	WIND:	10-15 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	20:08	ENGINE OFF:	2:07
ENGINE TIME:			05:59
TAKEOFF:	20:33	LANDING:	1:56
AIR TIME			05:23

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
027_FL6	3106	23:50:02	23:51:50	400	300 10	
028_FL5	3105	23:53:26	23:55:08	400	300 10	
029_FL4	3104	23:56:59	23:58:24	400	300 10	
030_FL3	3103	00:00:14	00:01:34	400	300 10	
031_FL2	3102	00:03:47	00:04:57	400	300 10	
		00:07:00				Moving from BL31 to BL32
032_FL1	3101	00:07:07	00:08:07	400	300 10	
		00:10:13				DS: BL32_20220523_001013
000_FL19	3219	00:10:13	00:13:05	400	300 10	
001_FL18	3218	00:16:05	00:20:06	400	300 10	
002_FL17	3217	00:21:58	00:26:11	400	300 10	
003_FL16	3216	00:28:14	00:32:40	400	300 10	
004_FL15	3215	00:34:21	00:38:53	400	300 10	
005_FL14	3214	00:40:25	00:44:44	400	300 10	
006_FL13	3213	00:46:43	00:51:16	400	300 10	
007_FL12	3212	00:53:08	00:57:21	400	300 10	
008_FL11	3211	00:59:18	01:03:29	400	300 10	
009_FL10	3210	01:05:17	01:09:18	400	300 10	
010_FL9	3209	01:11:09	01:15:12	400	300 10	
011_FL8	3208	01:24:05	01:27:37	400	300 10	
012_FL7	3207	01:29:23	01:32:32	400	300 10	
013_FL6	3206	01:34:27	01:37:08	400	300 10	
014_FL5	3205	01:38:44	01:40:58	400	300 10	
015_FL4	3204	01:48:51	01:50:28	400	300 10	
		01:56:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL30, QC61	DATE:	24 May 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 1500ft
BASE STATION:	TBU1	WIND:	10 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	19:25	ENGINE OFF:	21:42
TAKEOFF:	19:45	LANDING:	21:23
		ENGINE TIME:	02:17
		AIR TIME	01:38

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:45:00				Takeoff
		19:50:27				DS: QC61_20220524_195027
000_FL1	6101	19:50:27	19:51:54	400	300 10	BAD:
		19:57:37				DS: BL30_20220524_195737
000_FL1	3001	19:57:37	20:00:42	400	300 10	BAD: Eyesafety due to terrain
001_FL1	3001	20:01:29	20:01:36	400	300 10	BAD: Eyesafety due to terrain
002_FL2	3002	20:05:22	20:07:36	400	300 10	BAD:
003_FL5	3005	20:12:34	20:18:51	400	300 10	
004_FL6	3006	20:21:11	20:27:29	400	300 10	
005_FL7	3007	20:29:33	20:35:52	400	300 10	
006_FL8	3008	20:37:53	20:44:04	400	300 10	
007_FL9	3009	20:46:17	20:52:27	400	300 10	
008_FL10	3010	20:54:41	21:00:38	400	300 10	
		21:01:00				Reached extinction
009_FL4	3004	21:03:03	21:07:05	400	300 10	BAD:
		21:17:00				Ended survey, terrain causing eyesafety
		21:23:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL30, BL31, BL32, QC61	DATE:	2 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 1500ft
BASE STATION:	TBU1	WIND:	10 kts @ 50
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	19:40	ENGINE OFF:	23:59
TAKEOFF:	19:59	LANDING:	23:39
		ENGINE TIME:	04:19
		AIR TIME	03:40

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:59:00				Takeoff
		20:06:31				DS: BL31_20220602_200631
000_FL51	3150	20:06:31	20:07:28	400	300 10	BAD: Laser was not firing
001_FL51	3150	20:10:11	20:11:08	400	300 10	
002_FL50	3149	20:13:08	20:14:08	400	300 10	
003_FL49	3148	20:16:54	20:17:55	400	300 10	
004_FL48	3147	20:20:23	20:21:26	400	300 10	
005_FL47	3146	20:23:16	20:24:16	400	300 10	
006_FL46	3145	20:26:15	20:27:55	400	300 10	
007_FL45	3144	20:30:37	20:32:26	400	300 10	
008_FL44	3143	20:35:04	20:36:07	400	300 10	
009_FL43	3142	20:38:48	20:39:51	400	300 10	
010_FL42	3141	20:42:58	20:45:04	400	300 10	
011_FL41	3140	20:47:10	20:49:06	400	300 10	
012_FL40	3139	20:52:39	20:53:35	400	300 10	
013_FL39	3138	21:10:29	21:11:26	400	300 10	
014_FL38	3137	21:13:57	21:14:58	400	300 10	
015_FL37	3136	21:17:22	21:18:25	400	300 10	
016_FL36	3135	21:20:44	21:22:12	400	300 10	
017_FL35	3134	21:24:24	21:25:48	400	300 10	
		21:28:17				DS: BL32_20220602_212817
000_FL38	3237	21:28:17	21:29:07	400	300 10	
001_FL37	3236	21:31:20	21:32:51	400	300 10	
002_FL36	3235	21:35:13	21:36:47	400	300 10	
003_FL35	3234	21:38:35	21:40:04	400	300 10	
004_FL34	3233	21:42:45	21:44:25	400	300 10	
005_FL33	3232	21:46:34	21:48:12	400	300 10	
006_FL32	3231	21:50:38	21:52:22	400	300 10	
007_FL31	3230	21:54:01	21:54:56	400	300 10	
008_FL30	3229	21:56:59	21:57:50	400	300 10	
009_FL29	3228	21:59:52	22:00:47	400	300 10	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL30, BL31, BL32, QC61	DATE:	2 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.

MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 1500ft
BASE STATION:	TBU1	WIND:	10 kts @ 50
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04

ENGINE START:	19:40	ENGINE OFF:	23:59	ENGINE TIME:	04:19
TAKEOFF:	19:59	LANDING:	23:39	AIR TIME	03:40

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
010_FL28	3227	22:03:10	22:04:27	400	300 10	
011_FL27	3226	22:06:20	22:07:49	400	300 10	
012_FL26	3225	22:10:18	22:11:46	400	300 10	
013_FL25	3224	22:13:50	22:15:03	400	300 10	
014_FL24	3223	22:17:30	22:18:43	400	300 10	
015_FL23	3222	22:20:54	22:22:06	400	300 10	
016_FL22	3221	22:24:36	22:25:45	400	300 10	
017_FL21	3220	22:27:53	22:28:59	400	300 10	
		22:31:21				DS: BL30_20220602_223121
000_FL24	3023	22:31:21	22:32:26	400	300 10	
001_FL23	3022	22:51:18	22:51:30	400	300 10	BAD: Eyesafety
002_FL23	3022	22:52:33	22:52:50	400	300 10	BAD: Eyesafety
003_FL23	3022	22:53:06	22:53:21	400	300 10	BAD: Eyesafety
004_FL23	3022	22:54:04	22:55:21	400	300 10	
005_FL22	3021	22:59:14	23:00:14	400	300 10	
006_FL21	3020	23:02:46	23:03:43	400	300 10	
007_FL20	3019	23:06:13	23:07:17	400	300 10	
008_FL19	3018	23:10:52	23:11:53	400	300 10	
009_FL18	3017	23:14:05	23:14:55	400	300 10	
		23:16:00				Ended survey
		23:32:28				DS: QC61_20220602_233228
000_FL1	6101	23:32:28	23:33:56	400	300 10	
		23:39:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL33, BL34, QC61	DATE:	6 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 2500ft
BASE STATION:	TBU1	WIND:	15 kts @ 160
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	19:30	ENGINE OFF:	0:08
TAKEOFF:	19:56	LANDING:	23:49
		ENGINE TIME:	04:38
		AIR TIME	03:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		19:56:00				Takeoff
		20:05:13				DS: BL34_20220606_200513
000_FL6	3406	20:05:13	20:07:34	620	440 14	
001_FL5	3405	20:10:11	20:13:04	620	440 14	
002_FL4	3404	20:15:41	20:18:38	620	440 14	
003_FL3	3403	20:20:55	20:23:51	620	440 14	
004_FL2	3402	20:26:18	20:29:24	620	440 14	
005_FL1	3401	20:35:13	20:37:46	620	440 14	
		20:41:54				DS: BL33_20220606_204154
000_FL27	3327	20:41:54	20:47:23	620	440 14	
001_FL26	3326	20:49:23	20:54:58	620	440 14	
002_FL25	3325	20:57:31	21:03:11	620	440 14	
003_FL24	3324	21:05:12	21:10:57	620	440 14	
004_FL23	3323	21:13:15	21:16:13	620	440 14	BAD: Eyesafety due to terrain
005_FL23	3323	21:20:33	21:26:19	620	440 13	
006_FL22	3322	21:28:16	21:33:58	620	440 13	
007_FL21	3321	21:35:35	21:41:10	620	440 13	
008_FL20	3320	21:43:50	21:49:20	620	440 13	
009_FL19	3319	21:51:09	21:56:31	620	440 13	
010_FL18	3318	21:59:03	22:04:06	620	440 13	
011_FL17	3317	22:05:40	22:10:36	620	440 13	
012_FL16	3316	22:12:53	22:17:27	620	440 13	
013_FL15	3315	22:18:59	22:23:21	620	440 13	
014_FL14	3314	22:25:44	22:29:53	620	440 13	
015_FL13	3313	22:31:37	22:35:37	620	440 13	
016_FL12	3312	22:37:35	22:41:13	620	440 13	
017_FL11	3311	22:42:58	22:46:25	620	440 13	
018_FL10	3310	22:48:25	22:51:41	620	440 13	
019_FL9	3309	22:53:26	22:56:44	620	440 14	
020_FL8	3308	22:58:42	23:01:50	620	440 14	
021_FL7	3307	23:03:33	23:06:43	620	440 14	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL33, BL34, QC61	DATE:	6 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 2500ft
BASE STATION:	TBU1	WIND:	15 kts @ 160
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	19:30	ENGINE OFF:	0:08
TAKEOFF:	19:56	LANDING:	23:49
		ENGINE TIME:	04:38
		AIR TIME	03:53

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
022_FL6	3306	23:08:41	23:11:32	620	440	14	
023_FL5	3305	23:13:18	23:16:04	620	440	14	
024_FL4	3304	23:18:22	23:20:39	620	440	14	
025_FL3	3303	23:27:50	23:30:01	620	440	14	
026_FL2	3302	23:32:10	23:34:08	620	440	14	
027_FL1	3301	23:35:53	23:37:42	620	440	14	
		23:42:34					DS: QC61_20220606_234234
000_FL1	6101	23:42:34	23:44:04	400	300	10	
		23:49:00					Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Fua'amotu (TBU)
LOCATION / AREA:	Tonga / BL33, BL34, QC61	DATE:	7 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.

MISSION ID:	Eua, Tonga	CLOUDS:	Clouds @ 2500ft
BASE STATION:	TBU1	WIND:	15-20 kts @ 120
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05

ENGINE START:	20:42	ENGINE OFF:	23:46	ENGINE TIME:	03:04
TAKEOFF:	21:10	LANDING:	23:26	AIR TIME	02:16

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		21:10:00				Takeoff
		21:16:20				DS: BL34_20220607_211620
000_FL7	3407	21:16:20	21:17:52	620	440 14	
001_FL8	3408	21:19:49	21:21:22	620	440 14	
002_FL9	3409	21:23:51	21:25:31	620	440 14	
		21:34:00				Soft Restart due to Sensor issues
		21:42:53				DS: BL34_20220607_214253
000_FL10	3410	21:42:53	21:45:12	620	440 14	
001_FL11	3411	21:47:37	21:49:57	620	440 14	Light rain on half of line
002_FL12	3412	21:52:01	21:54:23	620	440 14	Light rain at end of line
003_FL13	3413	21:56:38	21:58:55	620	440 14	Light rain at start of line
004_FL14	3414	22:00:49	22:01:53	620	440 14	
005_FL15	3415	22:04:03	22:05:30	620	440 14	
006_FL16	3416	22:07:16	22:08:56	620	440 14	
007_FL17	3417	22:11:02	22:12:32	620	440 14	
008_FL18	3418	22:14:54	22:16:10	620	440 14	
009_FL19	3419	22:18:21	22:19:35	620	440 14	
010_FL20	3420	22:22:38	22:23:51	620	440 14	
		22:43:05				DS: BL33_20220607_224305
000_FL28	3328	22:43:05	22:44:15	620	440 14	
001_FL29	3329	22:46:27	22:47:45	620	440 14	Light rain on line
002_FL30	3330	22:49:43	22:51:38	620	440 14	Light rain on line
003_FL31	3331	22:53:31	22:55:24	620	440 14	Light rain on first half of line
004_FL32	3332	22:57:05	22:59:14	620	440 14	
005_FL33	3333	23:01:22	23:03:49	620	440 14	
006_FL34	3334	23:05:37	23:08:12	620	440 14	Light rain at start of line
007_FL35	3335	23:10:17	23:12:49	620	440 14	Light rain at end of line
008_FL36	3336	23:14:22	23:15:26	620	440 14	
		23:20:36				DS: QC61_20220607_232036
000_FL1	6101	23:20:36	23:22:05	400	300 10	
		23:26:00				Landing

PROJECT NAME: 2021-81155 SPC Tonga
LOCATION / AREA: Tonga / BL50
AIRCRAFT: ZK-XLF
SYSTEM: Hawkeye 4X

BASE AIRPORT: Vava'u (VAV)
DATE: 9 June 2022
PILOT: Cole B.
OPERATOR: Noah K.

MISSION ID: Vavau
BASE STATION: VAV1
LIDAR DRIVE: HE4X-05

CLOUDS: Clouds @ 1800ft
WIND: 10-15 kts @ 90
RCD DRIVE: RCD-03

ENGINE START: 21:56 **ENGINE OFF:** 3:34 **ENGINE TIME:** 05:38
TAKEOFF: 22:11 **LANDING:** 3:14 **AIR TIME:** 05:03

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		22:11:00					Takeoff
		22:15:30					DS: BL50_20220609_221530
000_FL71	5071	22:15:30	22:16:20	400	300	10	BAD: Eyesafety due to terrain
001_FL71	5071	22:20:56	22:21:53	400	300	9	BAD: Eyesafety due to terrain
002_FL72	5095	22:28:20	22:35:06	400	300	9	
		22:37:00					Digitizer settings changed
003_FL71	5071	22:41:58	22:52:11	400	300	9	
004_FL70	5070	22:54:28	23:04:46	400	300	9	
005_FL69	5069	23:07:05	23:17:16	400	300	9	
006_FL68	5068	23:19:37	23:29:54	400	300	9	
007_FL67	5067	23:31:49	23:42:04	400	300	9	
008_FL66	5066	23:43:48	23:54:02	400	300	9	
009_FL65	5065	00:00:52	00:11:01	400	300	9	
010_FL64	5064	00:12:57	00:23:08	400	300	9	
011_FL63	5063	00:24:44	00:34:50	400	300	9	
012_FL62	5062	00:36:45	00:46:54	400	300	9	
013_FL61	5061	00:48:33	00:58:44	400	300	9	
014_FL60	5060	01:00:41	01:10:49	400	300	9	
015_FL59	5059	01:12:45	01:22:48	400	300	9	
016_FL58	5058	01:28:33	01:38:39	400	300	9	
017_FL57	5057	01:40:29	01:50:21	400	300	9	
018_FL56	5056	01:52:05	02:01:55	400	300	9	
019_FL55	5055	02:03:53	02:13:47	400	300	9	
020_FL54	5054	02:15:39	02:25:32	400	300	9	
021_FL53	5053	02:27:22	02:37:01	400	300	9	
022_FL52	5052	02:39:02	02:48:38	400	300	9	
023_FL51	5051	02:50:28	02:59:54	400	300	9	
024_FL50	5050	03:01:41	03:11:09	400	300	9	
		03:14:00					Landing

PROJECT NAME: 2021-81155 SPC Tonga
LOCATION / AREA: Tonga / BL50, QC61
AIRCRAFT: ZK-XLF
SYSTEM: Hawkeye 4X

BASE AIRPORT: Vava'u (VAV)
DATE: 10 June 2022
PILOT: Cole B.
OPERATOR: Noah K.

MISSION ID: Tonga, Vavau
BASE STATION: VAV1
LIDAR DRIVE: HE4X-06

CLOUDS: Clouds @ 1800ft
WIND: 15-20 kts @ 90
RCD DRIVE: RCD-05

ENGINE START: 20:11 **ENGINE OFF:** 1:29 **ENGINE TIME:** 05:18
TAKEOFF: 20:28 **LANDING:** 1:10 **AIR TIME** 04:42

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		20:28:00					Takeoff
		20:31:43					DS: QC61_20220610_203143
000_FL2	6102	20:31:43	20:33:13	400	300	9	
		20:37:12					DS: BL50_20220610_203712
000_FL49	5049	20:37:12	20:46:26	400	300	9	Light rain on line halfway through
001_FL48	5048	20:48:44	20:58:03	400	300	9	
002_FL47	5047	20:59:57	21:09:08	400	300	9	
003_FL46	5046	21:12:22	21:21:30	400	300	9	
004_FL45	5045	21:23:15	21:32:13	400	300	9	
005_FL44	5044	21:34:36	00:00:00	400		9	BAD: System Error: Could not pause laser
006_FL43	5043	21:45:12	00:00:00	400		9	BAD: Collection within turn
		21:52:00					Soft restart
		22:00:51					DS: BL50_20220610_220051
000_FL43	5043	22:00:51	22:09:39	400	300	9	
001_FL42	5042	22:11:51	22:20:46	400	300	9	
002_FL41	5041	22:22:42	22:31:15	400	300	9	
003_FL40	5040	22:33:24	22:42:09	400	300	9	
004_FL39	5039	22:44:17	22:52:45	400	300	9	
005_FL38	5038	22:54:38	23:03:16	400	300	9	
006_FL37	5037	23:05:33	23:14:04	400	300	9	
007_FL36	5036	23:16:14	23:24:45	400	300	9	
008_FL35	5035	23:26:44	23:35:15	400	300	9	
009_FL34	5034	23:37:11	23:45:48	400	300	9	
010_FL33	5033	23:47:38	23:56:05	400	300	9	
011_FL32	5032	23:58:37	00:07:04	400	300	9	
012_FL31	5031	00:14:28	00:22:40	400	300	9	
013_FL30	5030	00:24:40	00:31:56	400	300	9	Light rain on line toward the end
014_FL29	5029	00:34:06	00:41:07	400	300	9	
015_FL28	5028	00:43:02	00:50:08	400	300	9	Light rain at end of line
016_FL27	5027	00:52:07	00:59:00	400	300	9	Light rain at start of line
017_FL1	5001	01:02:18	01:04:12	400	300	9	Rain on line

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL50, QC61	DATE:	10 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1800ft
BASE STATION:	VAV1	WIND:	15-20 kts @ 90
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	20:11	ENGINE OFF:	1:29
TAKEOFF:	20:28	LANDING:	1:10
		ENGINE TIME:	05:18
		AIR TIME	04:42

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:05:00				Ending survey due to rain
		01:10:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL50, BL51, QC61	DATE:	12 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1700ft
BASE STATION:	VAV1	WIND:	15 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	20:00	ENGINE OFF:	2:12
TAKEOFF:	20:11	LANDING:	1:55
		ENGINE TIME:	06:12
		AIR TIME	05:44

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:11:00				Takeoff
		20:20:49				DS: QC61_20220612_202049
000_FL2	6102	20:20:49	20:22:26	400	300 9	
		20:26:57				DS: BL50_20220612_202657
000_FL44	5044	20:26:57	20:35:56	400	300 9	
001_FL26	5026	20:37:53	20:44:50	400	300 9	
002_FL25	5025	20:46:57	20:53:53	400	300 9	
003_FL24	5024	20:55:54	21:02:57	400	300 9	
004_FL23	5023	21:04:51	21:11:53	400	300 9	
005_FL22	5022	21:19:58	21:27:01	400	300 9	
006_FL21	5021	21:28:38	21:35:41	400	300 9	
007_FL20	5020	21:37:52	21:44:59	400	300 9	
008_FL19	5019	21:46:51	21:53:53	400	300 9	
009_FL18	5018	21:55:51	22:02:52	400	300 9	
010_FL17	5017	22:04:28	22:11:25	400	300 9	
011_FL16	5016	22:13:19	22:19:58	400	300 9	
012_FL15	5015	22:21:50	22:28:21	400	300 9	
013_FL14	5014	22:30:30	22:35:43	400	300 9	
014_FL13	5013	22:37:33	22:42:31	400	300 9	
015_FL12	5012	22:44:25	22:49:19	400	300 9	
016_FL11	5011	22:51:09	22:55:48	400	300 9	
017_FL10	5010	22:57:46	23:02:15	400	300 9	
018_FL9	5009	23:04:04	23:08:24	400	300 9	
019_FL8	5008	23:10:25	23:14:34	400	300 9	
020_FL7	5007	23:16:19	23:20:18	400	300 9	
021_FL6	5006	23:22:09	23:26:03	400	300 9	
022_FL5	5005	23:28:05	23:31:37	400	300 9	
023_FL4	5004	23:33:29	23:36:49	400	300 9	
024_FL3	5003	23:38:45	23:42:02	400	300 9	
025_FL2	5002	23:44:25	23:47:00	400	300 9	
026_FL1	5001	23:48:52	23:50:47	400	300 9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL50, BL51, QC61	DATE:	12 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1700ft
BASE STATION:	VAV1	WIND:	15 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	20:00	ENGINE OFF:	2:12
TAKEOFF:	20:11	LANDING:	1:55
		ENGINE TIME:	06:12
		AIR TIME	05:44

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
027_FL73	5096	23:53:02	00:00:30	400	300	9	
		00:05:34					DS: BL51_20220613_000534
000_FL1	5101	00:05:34	00:15:45	400	300	9	
001_FL2	5102	00:18:23	00:28:34	400	300	9	
002_FL3	5103	00:30:20	00:40:39	400	300	9	
003_FL4	5104	00:42:03	00:52:11	400	300	9	
004_FL5	5105	00:54:15	01:04:37	400	300	9	
005_FL6	5106	01:06:01	01:16:15	400	300	9	
006_FL7	5107	01:17:46	01:28:04	400	300	9	
007_FL8	5108	01:29:34	01:39:43	400	300	9	
008_FL9	5109	01:41:39	01:51:48	400	300	9	
		01:55:00					Landing

PROJECT NAME: 2021-81155 SPC Tonga
LOCATION / AREA: Tonga / BL51, QC61
AIRCRAFT: ZK-XLF
SYSTEM: Hawkeye 4X

BASE AIRPORT: Vava'u (VAV)
DATE: 13 June 2022
PILOT: Cole B.
OPERATOR: Noah K.

MISSION ID: Tonga, Vavau
BASE STATION: VAV1
LIDAR DRIVE: HE4X-06

CLOUDS: Clouds @ 1500ft
WIND: 15 kts @ 100
RCD DRIVE: RCD-05

ENGINE START: 19:45 **ENGINE OFF:** 1:53 **ENGINE TIME:** 06:08
TAKEOFF: 20:00 **LANDING:** 1:34 **AIR TIME** 05:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
		20:00:00					Takeoff
		20:03:09					DS: QC61_20220613_200309
000_FL2	6102	20:03:09	20:04:43	400	300	9	
		20:08:47					DS: BL51_20220613_200847
000_FL10	5110	20:08:47	20:18:57	400	300	9	
001_FL11	5111	20:20:54	20:30:59	400	300	9	
002_FL12	5112	20:32:59	20:42:53	400	300	9	
003_FL13	5113	20:44:45	20:54:35	400	300	9	
004_FL14	5114	20:56:35	21:06:29	400	300	9	
005_FL15	5115	21:08:14	21:18:10	400	300	9	
006_FL16	5116	21:20:03	21:29:43	400	300	9	
007_FL17	5117	21:31:45	21:41:34	400	300	9	
008_FL18	5118	21:43:26	21:53:14	400	300	9	
009_FL19	5119	21:55:16	22:05:01	400	300	9	
010_FL20	5120	22:06:32	22:16:11	400	300	9	
011_FL21	5121	22:18:38	22:28:19	400	300	9	
012_FL22	5122	22:30:17	22:39:57	400	300	9	
013_FL23	5123	22:41:32	22:51:13	400	300	9	
014_FL24	5124	22:53:05	23:02:39	400	300	9	
015_FL25	5125	23:04:16	23:13:59	400	300	9	
016_FL26	5126	23:15:31	23:25:41	400	300	9	BAD: AHAB Software froze on line
		23:26:00					Soft restart, AHAB software froze
		23:32:10					DS: BL51_20220613_233210
000_FL26	5126	23:32:10	23:41:48	400	300	9	
001_FL27	5127	23:43:32	23:53:06	400	300	9	
002_FL28	5128	23:54:49	00:04:36	400	300	9	
003_FL29	5129	00:05:59	00:15:37	400	300	9	
004_FL30	5130	00:17:48	00:26:32	400	300	9	
005_FL31	5131	00:27:57	00:36:36	400	300	9	
006_FL32	5132	00:38:14	00:47:02	400	300	9	
007_FL33	5133	00:48:28	00:57:17	400	300	9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL51, QC61	DATE:	13 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	15 kts @ 100
LIDAR DRIVE:	HE4X-06	RCD DRIVE:	RCD-05
ENGINE START:	19:45	ENGINE OFF:	1:53
TAKEOFF:	20:00	LANDING:	1:34
		ENGINE TIME:	06:08
		AIR TIME	05:34

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
008_FL34	5134	00:59:21	01:08:14	400	300	9	
009_FL35	5135	01:09:48	01:18:44	400	300	9	
010_FL36	5136	01:20:09	01:29:07	400	300	9	
		01:34:00					Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL51, QC61	DATE:	14 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	15 kts @ 90
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-03
ENGINE START:	19:50	ENGINE OFF:	23:50
TAKEOFF:	20:06	LANDING:	23:32
		ENGINE TIME:	04:00
		AIR TIME	03:26

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:06:00				Takeoff
		20:09:04				DS: QC61_20220614_200904
000_FL2	6102	20:09:04	20:10:33	400	300 9	
		20:14:19				DS: BL51_20220614_201419
000_FL37	5137	20:14:19	20:23:30	400	300 9	
001_FL38	5138	20:25:31	20:34:40	400	300 9	
002_FL39	5139	20:36:42	20:45:32	400	300 9	
003_FL40	5140	20:47:41	20:56:35	400	300 9	
004_FL41	5141	20:58:31	21:07:17	400	300 9	
005_FL42	5142	21:09:17	21:18:06	400	300 9	
006_FL43	5143	21:19:44	21:28:25	400	300 9	
007_FL44	5144	21:30:22	21:38:59	400	300 9	
008_FL45	5145	21:40:46	21:49:25	400	300 9	
009_FL46	5146	21:51:19	21:59:56	400	300 9	
010_FL47	5147	22:01:36	22:10:04	400	300 9	
011_FL48	5148	22:12:06	22:18:47	400	300 9	
012_FL49	5149	22:20:36	22:27:07	400	300 9	
013_FL50	5150	22:28:45	22:35:22	400	300 9	
014_FL51	5151	22:37:19	22:43:36	400	300 9	
015_FL52	5152	22:45:32	22:51:37	400	300 9	
016_FL53	5153	22:53:17	22:59:01	400	300 9	
017_FL54	5154	23:00:51	23:06:10	400	300 9	
018_FL55	5155	23:08:10	23:12:49	400	300 9	
019_FL56	5156	23:15:18	23:16:35	400	300 9	
020_FL57	5157	23:18:19	23:19:26	400	300 9	
021_FL58	5158	23:21:32	23:22:38	400	300 9	
		23:32:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL50, QC61	DATE:	19 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	15 kts @ 10
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-04
ENGINE START:	20:20	ENGINE OFF:	22:47
TAKEOFF:	20:37	LANDING:	22:35
		ENGINE TIME:	02:27
		AIR TIME	01:58

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:37:00				Takeoff
		20:37:00				Deep sensor pwr settings: 100 narrow, 150
		20:39:00				Soft restart because sensor wouldn't arm
		20:46:15				DS: QC61_20220619_204615
000_FL2	6102	20:46:15	20:47:50	400	300 9	
		20:52:11				DS: BL50_20220619_205211
000_FL7	5007	20:52:11	20:56:14	400	300 9	
001_FL8	5008	20:58:02	21:02:09	400	300 9	
002_FL9	5009	21:04:14	21:08:38	400	300 9	
		21:10:00				Deep sensor pwr settings: 150 narrow, 200
003_FL46	5046	21:11:35	21:20:46	400	300 9	
004_FL47	5047	21:29:26	21:38:40	400	300 9	
005_FL48	5048	21:40:29	21:49:49	400	300 9	
		21:51:00				Deep sensor pwr settings: 200 narrow, 250
006_FL58	5058	21:51:38	22:01:38	400	300 9	
007_FL59	5059	22:04:04	22:14:01	400	300 9	
008_FL71	5071	22:16:09	22:26:31	400	300 9	
		22:35:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL45, QC61	DATE:	21 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafahi, Tonga	CLOUDS:	Clouds @ 2000ft
BASE STATION:	VAV1	WIND:	10 kts @ 30
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	20:25	ENGINE OFF:	2:05
TAKEOFF:	20:50	LANDING:	1:44
		ENGINE TIME:	05:40
		AIR TIME	04:54

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:50:00				Takeoff
		20:50:00				Deep sensor pwr settings: 240 narrow, 280
		20:53:26				DS: QC61_20220621_205326
000_FL2	6102	20:53:26	20:54:59	400	300 9	BAD: Topo Laser never started
001_FL2	6102	20:56:51	20:58:17	400	300 9	
		21:10:00				Deep sensor pwr settings: 150 narrow, 200
		21:52:08				DS: BL45_20220621_215208
000_FL6	4506	21:52:08	21:52:43	620	300 9	BAD: Eyesafety on line
		21:56:49				DS: BL40_20220621_215649
000_FL1	4001	21:56:49	21:59:42	400	300 9	
		22:00:00				Deep sensor pwr settings: 200 narrow, 250
001_FL2	4002	22:01:28	22:04:29	400	300 9	
002_FL3	4003	22:06:22	22:09:31	400	300 9	
003_FL4	4004	22:11:11	22:14:25	400	300 9	
004_FL5	4005	22:16:11	22:19:24	400	300 9	
		22:20:00				Deep sensor pwr settings: 100 narrow, 150
005_FL6	4006	22:21:04	22:24:16	400	300 9	
006_FL7	4007	22:25:56	22:29:03	400	300 9	
007_FL8	4008	22:30:55	22:33:59	400	300 9	
008_FL9	4009	22:35:40	22:38:43	400	300 9	
		22:39:00				Deep sensor pwr settings: 50 narrow, 100
009_FL10	4010	22:40:28	22:43:33	400	300 9	
010_FL11	4011	22:45:01	22:48:03	400	300 9	
011_FL12	4012	22:49:50	22:52:56	400	300 9	
012_FL13	4013	22:54:25	22:57:25	400	300 9	
013_FL14	4014	22:59:10	23:02:12	400	300 9	
014_FL15	4015	23:03:57	23:06:57	400	300 9	
015_FL16	4016	23:08:37	23:11:30	400	300 9	
016_FL17	4017	23:13:07	23:16:03	400	300 9	
017_FL18	4018	23:17:51	23:20:38	400	300 9	
018_FL19	4019	23:22:24	23:25:14	400	300 9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL45, QC61	DATE:	21 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.

MISSION ID:	Tafahi, Tonga	CLOUDS:	Clouds @ 2000ft
BASE STATION:	VAV1	WIND:	10 kts @ 30
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03

ENGINE START:	20:25	ENGINE OFF:	2:05	ENGINE TIME:	05:40
TAKEOFF:	20:50	LANDING:	1:44	AIR TIME	04:54

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO		REMARKS
					PRF	PWR	
019_FL20	4020	23:26:55	23:29:44	400	300	9	
020_FL21	4021	23:31:16	23:33:56	400	300	9	
021_FL22	4022	23:35:32	23:38:11	400	300	9	
022_FL23	4023	23:39:36	23:42:11	400	300	9	
023_FL24	4024	23:44:03	23:46:37	400	300	9	
024_FL25	4025	23:48:21	23:50:47	400	300	9	
025_FL26	4026	23:52:49	23:55:09	400	300	9	
026_FL27	4027	23:56:47	23:59:04	400	300	9	
027_FL28	4028	00:00:57	00:03:12	400	300	9	
028_FL29	4029	00:04:46	00:06:55	400	300	9	
029_FL30	4030	00:08:44	00:10:53	400	300	9	
030_FL31	4031	00:12:14	00:14:12	400	300	9	
031_FL32	4032	00:15:57	00:17:52	400	300	9	
032_FL33	4033	00:19:39	00:21:29	400	300	9	
033_FL34	4034	00:23:23	00:25:13	400	300	9	
034_FL35	4035	00:26:52	00:28:31	400	300	9	
035_FL36	4036	00:30:20	00:31:53	400	300	9	
036_FL37	4037	00:33:37	00:35:00	400	300	9	
037_FL38	4038	00:37:16	00:38:28	400	300	9	
038_FL39	4039	00:40:12	00:41:12	400	300	9	
039_FL40	4040	00:43:26	00:44:15	400	300	9	
040_FL41	4095	00:46:33	00:49:28	400	300	9	
		01:42:59					DS: QC61_20220622_014259
000_FL2	6102	01:42:59	01:44:27	400	300	9	
		01:44:00					Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL41, BL42, BL44	DATE:	27 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafari	CLOUDS:	Clouds @ 1500ft
BASE STATION:	PPP	WIND:	20 kts @ 90
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-05
ENGINE START:	21:57	ENGINE OFF:	1:31
TAKEOFF:	22:13	LANDING:	1:15
		ENGINE TIME:	03:34
		AIR TIME	03:02

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		22:13:00				Takeoff
		22:13:00				Deep sensor pwr settings: 240 narrow, 28
		23:08:17				DS: BL41_20220627_230817
000_FL1	4101	23:08:17	23:09:41	400	300 9	
001_FL2	4102	23:11:59	23:13:34	400	300 9	
002_FL3	4103	23:15:36	23:17:10	400	300 9	
003_FL4	4104	23:18:49	23:20:21	400	300 9	
004_FL5	4105	23:22:13	23:23:46	400	300 9	
005_FL6	4106	23:25:58	23:27:29	400	300 9	
006_FL7	4107	23:29:22	23:30:54	400	300 9	
007_FL8	4108	23:33:03	23:34:34	400	300 9	
008_FL9	4109	23:36:36	23:38:06	400	300 9	
		23:40:37				DS: BL44_20220627_234037
000_FL5	4405	23:40:37	23:42:22	400	300 9	
001_FL4	4404	23:44:21	23:45:59	400	300 9	
002_FL3	4403	23:47:52	23:49:31	400	300 9	
003_FL2	4402	23:51:26	23:53:03	400	300 9	
004_FL1	4401	23:55:06	23:56:30	400	300 9	
		23:59:57				DS: BL42_20220627_235957
000_FL6	4206	23:59:57	00:01:00	400	300 9	
001_FL5	4205	00:02:57	00:04:14	400	300 9	
002_FL4	4204	00:06:19	00:07:32	400	300 9	
003_FL3	4203	00:09:33	00:10:57	400	300 9	
004_FL2	4202	00:13:07	00:14:30	400	300 9	
005_FL1	4201	00:16:22	00:00:00	400	9	BAD: No HE4X Files
		01:15:00				Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL52, QC61	DATE:	28 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	20 kts @ 60
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	21:30	ENGINE OFF:	2:07
TAKEOFF:	21:49	LANDING:	1:49
		ENGINE TIME:	04:37
		AIR TIME	04:00

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		01:27:00				Soft Restart, software stop
		21:49:00				Takeoff
		21:50:00				Soft Restart, software issues
		21:53:39				DS: QC61_20220628_215339
000_FL2	6102	21:53:39	21:55:08	400	500 9	
		21:58:10				DS: BL52_20220628_215810
000_FL9	5209	21:58:10	21:59:11	400	500 9	
001_FL99	5299	22:02:38	22:03:48	400	500 9	
002_FL98	5298	22:06:10	22:07:23	400	500 9	
003_FL97	5297	22:09:23	22:10:39	400	500 9	
004_FL93	5293	22:15:25	22:17:17	400	500 9	
005_FL92	5292	22:19:46	22:21:43	400	500 9	
006_FL91	5291	22:23:35	22:25:33	400	500 9	
007_FL90	5290	22:27:53	22:29:56	400	500 9	
008_FL89	5289	22:32:17	22:34:19	400	500 9	
009_FL88	5288	22:36:53	22:38:58	400	500 9	
010_FL87	5287	22:41:08	22:43:09	400	500 9	
011_FL86	5286	22:45:33	22:47:37	400	500 9	
012_FL96	5296	22:50:17	22:51:23	400	500 9	
013_FL95	5295	22:53:52	22:55:02	400	500 9	
014_FL94	5294	22:57:19	22:58:28	400	500 9	
015_FL8	5208	23:01:06	23:02:04	400	500 9	
016_FL77	5277	23:05:03	23:07:06	400	500 9	
017_FL79	5279	23:08:23	23:08:45	400	500 9	BAD: Eyesafety
018_FL32	5232	23:11:25	23:12:33	400	300 9	
019_FL33	5233	23:15:34	23:16:30	400	300 9	
020_FL16	5216	23:18:36	23:19:33	400	300 9	
021_FL71	5271	23:22:12	23:23:08	400	300 9	
022_FL70	5270	23:25:07	23:26:04	400	300 9	
023_FL69	5269	23:28:34	23:29:34	400	300 9	
024_FL68	5268	23:31:28	23:32:28	400	300 9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL52, QC61	DATE:	28 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	20 kts @ 60
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	21:30	ENGINE OFF:	2:07
TAKEOFF:	21:49	LANDING:	1:49
		ENGINE TIME:	04:37
		AIR TIME	04:00

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO		REMARKS
					PRF	PWR	
025_FL67	5267	23:35:06	23:36:04	400	300	9	
026_FL66	5266	23:38:21	23:39:16	400	300	9	
027_FL42	5242	23:43:00	23:44:02	400	300	9	
028_FL43	5243	23:46:34	23:47:32	400	300	9	
029_FL44	5244	23:49:31	23:50:39	400	300	9	
030_FL45	5245	23:53:02	23:53:51	400	300	9	
031_FL46	5246	23:55:50	23:56:52	400	300	9	
032_FL47	5247	23:59:20	00:00:18	400	300	9	
033_FL48	5248	00:02:23	00:03:29	400	300	9	
034_FL17	5217	00:05:56	00:06:44	400	300	9	
035_FL18	5218	00:08:55	00:09:46	400	300	9	
036_FL19	5219	00:12:02	00:12:59	400	300	9	
037_FL20	5220	00:14:53	00:15:52	400	300	9	
038_FL21	5221	00:18:00	00:18:53	400	300	9	
039_FL22	5222	00:21:00	00:22:12	400	300	9	
040_FL23	5223	00:24:37	00:25:41	400	300	9	
041_FL24	5224	00:27:47	00:28:58	400	300	9	
042_FL25	5225	00:31:27	00:32:31	400	300	9	
043_FL26	5226	00:34:34	00:35:42	400	300	9	
044_FL27	5227	00:37:39	00:38:42	400	300	9	
045_FL28	5228	00:40:43	00:41:51	400	300	9	
046_FL29	5229	00:44:38	00:45:40	400	300	9	
047_FL30	5230	00:47:24	00:48:28	400	300	9	
048_FL31	5231	00:50:32	00:51:19	400	300	9	
049_FL12	5212	00:53:51	00:54:43	400	300	9	
050_FL11	5211	00:56:57	00:57:54	400	300	9	
051_FL10	5210	00:59:39	01:00:35	400	300	9	
052_FL1	5201	01:04:00	01:05:02	400	300	9	
053_FL2	5202	01:07:06	01:07:57	400	300	9	
054_FL3	5203	01:10:29	01:11:26	400	300	9	
055_FL4	5204	01:13:29	01:14:18	400	300	9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL52, QC61	DATE:	28 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	20 kts @ 60
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-03
ENGINE START:	21:30	ENGINE OFF:	2:07
TAKEOFF:	21:49	LANDING:	1:49
		ENGINE TIME:	04:37
		AIR TIME	04:00

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
056_FL5	5205	01:16:14	01:17:16	400	300	9	
057_FL6	5206	01:19:37	01:20:37	400	300	9	
058_FL7	5207	01:22:52	01:23:49	400	300	9	BAD: Software stop
		01:31:34					DS: BL52_20220629_013134
000_FL7	5207	01:31:34	01:32:40	400	300	9	BAD: Eyesafety toward end of line
001_FL13	5213	01:34:57	01:35:59	400	300	9	
002_FL77	5277	01:39:17	01:41:21	400	500	9	
003_FL76	5276	01:43:08	01:45:01	400	500	9	BAD: Eyesafety
		01:49:00					Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL43, BL52, QC61	DATE:	29 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafahi, Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	15-20 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	16:49	ENGINE OFF:	21:45
TAKEOFF:	17:10	LANDING:	21:28
		ENGINE TIME:	04:56
		AIR TIME	04:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		17:10:00				Takeoff
		18:04:03				DS: BL43_20220629_180403
000_FL4	4304	18:04:03	18:05:15	400	300 9	
		18:09:16				DS: BL40_20220629_180916
000_FL45	4045	18:09:16	18:12:16	400	300 9	
001_FL54	4054	18:14:22	18:17:21	400	300 9	
002_FL53	4053	18:19:31	18:22:31	400	300 9	
003_FL52	4052	18:25:17	18:28:14	400	300 9	
004_FL51	4051	18:30:08	18:33:10	400	300 9	
005_FL50	4050	18:35:31	18:38:36	400	300 9	
006_FL49	4049	18:40:28	18:43:32	400	300 9	
007_FL48	4048	18:45:46	18:48:42	400	300 9	
008_FL47	4047	18:50:35	18:53:35	400	300 9	
009_FL41	4095	18:57:14	18:59:01	400	300 9	
010_FL42	4042	19:01:43	19:03:40	400	300 9	
011_FL43	4043	19:05:53	19:07:54	400	300 9	
012_FL44	4044	19:10:32	19:12:42	400	300 9	
013_FL46	4046	19:15:17	19:18:17	400	300 9	Light rain in middle of line
		20:09:05				DS: QC61_20220629_200905
000_FL2	6102	20:09:05	20:10:41	400	300 9	
		20:15:09				DS: BL52_20220629_201509
000_FL72	5272	20:15:09	20:16:17	400	500 9	
001_FL73	5273	20:19:02	20:20:49	400	500 9	
002_FL74	5274	20:22:50	20:24:42	400	500 9	
		20:28:00				Soft Restart, software stop
		20:37:08				DS: BL52_20220629_203708
000_FL85	5285	20:37:08	20:38:24	400	500 9	
001_FL84	5284	20:40:42	20:42:02	400	500 9	
002_FL83	5283	20:44:22	20:45:43	400	500 9	
003_FL82	5282	20:48:04	20:49:25	400	500 9	
004_FL81	5281	20:51:45	20:53:03	400	500 9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL43, BL52, QC61	DATE:	29 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafahi, Tonga, Vavau	CLOUDS:	Clouds @ 1500ft
BASE STATION:	VAV1	WIND:	15-20 kts @ 80
LIDAR DRIVE:	HE4X-05	RCD DRIVE:	RCD-04
ENGINE START:	16:49	ENGINE OFF:	21:45
TAKEOFF:	17:10	LANDING:	21:28
		ENGINE TIME:	04:56
		AIR TIME	04:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR		REMARKS
005_FL80	5280	20:55:08	20:56:18	400	500	9	
006_FL79	5279	20:58:31	20:59:31	400	500	9	
007_FL78	5278	21:01:47	21:02:47	400	500	9	
008_FL14	5214	21:05:26	21:06:27	400	300	9	
009_FL15	5215	21:09:03	21:09:50	400	300	9	
010_FL77	5277	21:14:09	21:16:04	400	500	9	
011_FL76	5276	21:18:37	21:20:34	400	500	9	
012_FL75	5275	21:22:20	21:24:13	400	500	9	BAD: Software Stop
		21:28:00					Landing

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL41, BL42, BL43, BL44, BL45, QC61	DATE:	30 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafahi, Tonga	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	20-25 kts @ 90
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	20:10	ENGINE OFF:	23:55
TAKEOFF:	20:26	LANDING:	23:44
		ENGINE TIME:	03:45
		AIR TIME	03:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		20:26:00				Takeoff
		21:18:50				DS: BL40_20220630_211850
000_FL56	4056	21:18:50	21:19:42	400	300 9	
001_FL55	4055	21:22:31	21:23:22	400	300 9	
		21:25:50				DS: BL45_20220630_212550
000_FL6	4506	21:25:50	21:26:06	620	440 9	BAD: Eye safety
001_FL6	4506	21:30:11	21:31:44	620	440 9	
002_FL5	4505	21:34:06	21:35:39	620	440 10	
003_FL4	4504	21:37:36	21:38:34	620	440 10	
		21:41:55				DS: BL44_20220630_214155
000_FL1	4401	21:41:55	21:43:18	400	300 9	
001_FL2	4402	21:45:29	21:47:09	400	300 9	
		21:50:27				DS: BL41_20220630_215027
000_FL9	4109	21:50:27	21:52:00	400	300 9	
001_FL8	4108	21:53:43	21:55:16	400	300 9	
002_FL7	4107	21:57:32	21:59:07	400	300 9	
003_FL6	4106	22:00:58	22:02:39	400	300 9	
004_FL5	4105	22:04:52	22:06:32	400	300 9	
005_FL4	4104	22:08:52	22:10:26	400	300 9	
006_FL3	4103	22:12:42	22:14:18	400	300 9	
007_FL2	4102	22:16:31	22:18:14	400	300 9	
008_FL1	4101	22:20:59	22:22:33	400	300 9	
		22:25:07				DS: BL43_20220630_222507
000_FL4	4304	22:25:07	22:26:20	400	300 9	
		22:28:03				DS: BL42_20220630_222803
000_FL6	4206	22:28:03	22:29:10	400	300 9	
001_FL5	4205	22:31:16	22:32:41	400	300 9	
002_FL4	4204	22:34:42	22:36:08	400	300 9	
003_FL3	4203	22:38:12	22:39:44	400	300 9	
004_FL2	4202	22:41:53	22:43:24	400	300 9	
005_FL1	4201	22:45:24	22:46:57	400	300 9	

PROJECT NAME:	2021-81155 SPC Tonga	BASE AIRPORT:	Vava'u (VAV)
LOCATION / AREA:	Tonga / BL40, BL41, BL42, BL43, BL44, BL45, QC61	DATE:	30 June 2022
AIRCRAFT:	ZK-XLF	PILOT:	Cole B.
SYSTEM:	Hawkeye 4X	OPERATOR:	Noah K.
MISSION ID:	Tafahi, Tonga	CLOUDS:	Clouds @ 2000ft
BASE STATION:	PPP	WIND:	20-25 kts @ 90
LIDAR DRIVE:	HE4X-07	RCD DRIVE:	RCD-05
ENGINE START:	20:10	ENGINE OFF:	23:55
TAKEOFF:	20:26	LANDING:	23:44
		ENGINE TIME:	03:45
		AIR TIME	03:18

FL #	LINE #	START TIME	END TIME	ALTITUDE	TOPO PRF PWR	REMARKS
		23:39:43				DS: QC61_20220630_233943
000_FL2	6102	23:39:43	23:41:18	400	300 9	
		23:44:00				Landing

APPENDIX B : PROCESSING LOGS

PROJECT NAME: 81155 SPC Tonga
LOCATION: Tonga
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

PID	Project Name	Latitude			Longitude			Height (m)	XY Datum	Z Datum	Source	Comments
		Deg	Min	Sec	Deg	Min	Sec					
TBU1	Tonga	S21	14	40.88798	W175	08	16.48436	85.703	ITRF2014 (Epoch 2022.38)	ITRF2014 (Epoch 2022.38)	RTX	Tonga Cal/'Eua Production
VAV1	Tonga	S18	35	08.08803	W173	58	03.37296	115.050	ITRF2014 (Epoch 2022.44)	ITRF2014 (Epoch 2022.44)	RTX	Vava'u Production

PROJECT NAME: 81155 SPC Tonga
LOCATION: Tonga
AIRCRAFT: Cessna Reims 406 (ZK-XLF)
SYSTEM: Hawkeye 4X

Project Name	Download Airborne Data	Create IE Project Directory	Copy Data to IE Project / Raw	Run Project Wizard	Base Station			Check Base Coordinate &	Check Lever Arms	Process Time Window		Process TC	Review QC Plots	Separation			Solution Status	Comments
					Station ID	Receiver Type	Antenna (ARP) Height (m)			Start Time (GPS Week Time)	End Time (GPS Week Time)			East RMS (m)	North RMS (m)	Up RMS (m)		
2022-05-20A_CH4X	GT	GT	GT	GT	TBU1	R10	1.8	GT	GT			GT	GT	0.013	0.016	0.035	Final	Calibration
2022-05-20A_HE4X	GT	GT	AS	AS	TBU1	R10	1.8	AS	AS			AS	AS	0.009	0.015	0.035	Final	Calibration
2022-05-21A_CH4X	GT	GT	GT	GT	TBU1	R10	1.8	GT	GT			GT	GT	0.009	0.024	0.060	Final	Eua Production
2022-05-21A_HE4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	AB	0.010	0.023	0.061	Final	Eua Production
2022-05-22A_CH4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	AB	0.018	0.015	0.034	Final	Eua Production
2022-05-22A_HE4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	AB	0.018	0.013	0.032	Final	Eua Production
2022-05-24A_CH4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	GT	0.046	0.058	0.137	Final	Eua Production
2022-05-24A_HE4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	GT	0.061	0.065	0.191	Final	Eua Production
2022-06-02A_CH4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	AB	0.009	0.012	0.042	Final	Eua Bathy Reflights
2022-06-02A_HE4X	AB	AB	AB	AB	TBU1	R10	1.8	AB	AB			AB	AB	0.009	0.011	0.046	Final	Eua Bathy Reflights
2022-06-03A_CH4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.027	0.006	0.049	Final	Eua Production
2022-06-03A_HE4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.015	0.016	0.050	Final	Eua Production
2022-06-06A_CH4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.038	0.036	0.050	Final	Eua Production
2022-06-06A_HE4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.041	0.042	0.066	Final	Eua Production
2022-06-07A_CH4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.014	0.019	0.050	Final	Eua Production
2022-06-07A_HE4X	NK	NK	NK	NK	TBU1	R10	1.8	NK	NK			NK	NK	0.015	0.020	0.041	Final	Eua Production
2022-06-09A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.019	0.008	0.031	Final	Vava'u Production
2022-06-09A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.017	0.008	0.030	Final	Vava'u Production
2022-06-10A_CH4X	NK	AB	AB	AB	VAV1	R10	1.8	AB	AB			AB	AB	0.016	0.048	0.098	Final	Multipass/Vava'u Production
2022-06-10A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.016	0.050	0.099	Final	Multipass/Vava'u Production
2022-06-12A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.041	0.016	0.031	Final	Vava'u Production
2022-06-12A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.044	0.016	0.027	Final	Vava'u Production
2022-06-13A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.045	0.088	0.098	Final	Multipass/Vava'u Production
2022-06-13A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.059	0.080	0.101	Final	Multipass/Vava'u Production
2022-06-14A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.017	0.018	0.050	Final	Vava'u Production
2022-06-14A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.020	0.028	0.046	Final	Vava'u Production
2022-06-19A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.014	0.004	0.021	Final	Vava'u Bathy Reflights
2022-06-19A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.012	0.006	0.027	Final	Vava'u Bathy Reflights
2022-06-21A_CH4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			NK	NK	0.025	0.017	0.173	Final	Tafari Production
2022-06-21A_HE4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			NK	NK	0.017	0.019	0.191	Final	Tafari Production
2022-06-27A_CH4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			NK	NK	0.026	0.019	0.083	Final	Tafari Production
2022-06-27A_HE4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			NK	NK	0.023	0.017	0.092	Final	Tafari Production
2022-06-28A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.016	0.006	0.031	Final	Vava'u Topo and Bathy Reflights
2022-06-28A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.017	0.007	0.027	Final	Vava'u Topo and Bathy Reflights
2022-06-29A_CH4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.010	0.011	0.039	Final	Vava'u and Tafari Reflights (Multipass)
2022-06-29A_HE4X	NK	NK	NK	NK	VAV1	R10	1.8	NK	NK			NK	NK	0.013	0.011	0.037	Final	Vava'u and Tafari Reflights (Multipass)
2022-06-30A_CH4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			AB	AB	0.019	0.017	0.131	Final	Tafari Reflights
2022-06-30A_HE4X	NK	NK	NK	NK	PPP	N/A	N/A	NK	NK			AB	AB	0.023	0.027	0.128	Final	Tafari Reflights